EX.NO: 14

IMPLEMENTATION OF PREDICTIVE MODEL IN R

AIM:

To implement a predictive model in R using linear regression to forecast outcomes based on given data.

PROCEDURE:

1. Load the Dataset – Use built-in or custom datasets for predictive modeling.
2. Split Data into Training and Testing Sets – Divide the dataset for model training and evaluation.
3. Build and Train the Model – Use the lm() function for linear regression or other machine learning models.
4. Make Predictions and Evaluate Performance – Use predict() to forecast values and assess model accuracy.

SOURCE CODE:

```
# Load dataset
data <- data.frame(
  Experience = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10),
  Salary = c(30000, 35000, 40000, 45000, 50000, 55000, 60000, 65000, 70000, 75000)
)

# Split data into training and testing sets
train_data <- data[1:8, ]  # First 8 rows for training
test_data <- data[9:10, ]  # Last 2 rows for testing

# Build Linear Regression Model
model <- lm(Salary ~ Experience, data = train_data)
```

```r
# Predict salary for test data
predictions <- predict(model, newdata = test_data)

# Display results
print("Predicted Salaries:")
print(predictions)

# Plot the regression line
plot(data$Experience, data$Salary, col="blue", pch=16, main="Experience vs Salary")
abline(model, col="red", lwd=2)
```

OUTPUT:

1. Predicted Salary Values – Prints the predicted salaries for the test data.
2. Regression Line Plot – Displays a scatter plot of experience vs. salary with a fitted regression line.

Result:

The predictive model was successfully implemented in R using linear regression, and future salary predictions based on experience were generated.